

- Q.32 Explain any one type of temperature transducers.
- Q.33 Explain the classification and performance characteristics of electrode and sensors.
- Q.34 Explain digital signal processing system with block diagram.
- Q.35 Explain classification and performance characteristics of electrodes and sensors

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain ECG & its measurements. Also Explain ECG machine working with block diagram.
- Q.37 Explain the basic component of biomedical recording system with block diagram.
- Q.38 Explain blood flow measurements using electromagnetic and ultrasonic method.

No. of Printed Pages : 4
Roll No.

183252

5th Sem / Med. Eltx
Subject:- Biomedical Sensors & Transducers (BMST)

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 LVDT is used to measure
- a) voltage b) displacement
- c) temperature d) None of the above
- Q.2 Which transducer best suited for brain waves
- a) ECG sensor b) EEG sensor
- c) EMG sensor d) All of the above
- Q.3 Which of the following is an active transducer
- a) Thermocouple b) Strain gauge
- c) Thermistor d) Photo-diode
- Q.4 Capacitive transducer are sensitive to change in
- a) Resistance b) Temperature
- c) Displacement d) Voltage
- Q.5 A biomedical signal is generally
- a) Pure DC b) High frequency AC
- c) Low frequency AC d) Random Noise

- Q.6 Which type of sensor used in spirometer.
 a) Pressure sensor b) Flow sensor
 c) Temperature sensor d) All of the above
- Q.7 EMG measures
 a) Brain activity b) Heart activity
 c) Muscle activity d) Eye movement
- Q.8 How many electrodes placed in 10/20 electrode placement system
 a) 19 b) 20
 c) 21 d) 10
- Q.9 A pulse oximeter primarily use for measurement
 a) Piezoelectric signal b) Pressure difference
 c) Light absorption d) Temperature change
- Q.10 Which is not a biosensor component
 a) Bio receptor b) Transducer
 c) Display d) Thermostat

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Biomedical sensor
 Q.12 Transducer
 Q.13 Give an example of temperature transducer
 Q.14 Photo-diode

- Q.15 Expand ECG
 Q.16 Name one non-invasive biomedical sensor
 Q.17 Draw normal waveform pattern of ECG
 Q.18 Expand LVDT
 Q.19 Seebeck effect
 Q.20 Example of chemical biosensor.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain any three types of biomedical electrode.
 Q.22 Explain classification of transducers.
 Q.23 Write a note thermocouple.
 Q.24 Write a note on signal conditioning.
 Q.25 Explain difference between EMG and EEG signals.
 Q.26 Explain importance of cardiovascular measurements.
 Q.27 Write a note on principle of photoelectric transducers.
 Q.28 Explain the applications of fibre optic sensors in medicine.
 Q.29 Explain principle of ultrasonic measurements.
 Q.30 What is the importance of differential amplifiers in biological signal conditioning.
 Q.31 Write a short note on electrode-electrolyte interface.